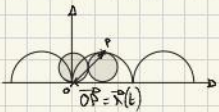


Lecture 6: Kepler's Second Law

$(x(t), y(t), z(t))$ = position of a moving point

Position vector: $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$



Example: cycloid (wheel rad. is 1, rotate at unit speed)
 $\vec{r}(t) = \langle t - \sin t, 1 - \cos t \rangle$

Velocity Vector: $\vec{v} = \frac{d\vec{r}}{dt} = \langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \rangle$

speed (scalar):

$$|\vec{v}| = \sqrt{(1 - \cos t)^2 + \sin^2 t} \\ = \sqrt{1 - 2\cos t + \cos^2 t + \sin^2 t} \\ = \sqrt{2 - 2\cos t}$$

Example for cycloid: $\vec{v} = \langle 1 - \cos t, \sin t \rangle$ (at $t=0$: $\vec{v}=0$)

Acceleration: a vector!

$\vec{a} = \frac{d\vec{v}}{dt}$ e.g. cycloid: $\vec{a} = \langle \sin t, \cos t \rangle$ at $t=0$: $\vec{a} = \langle 0, 1 \rangle$



Arc length: s = distance travelled along trajectory s versus t ?

$$\left| \frac{ds}{dt} \right| = \text{speed} = |\vec{v}|$$

Example: length of a cycloid: $s = \int_0^T \sqrt{2 - 2\cos t} dt$

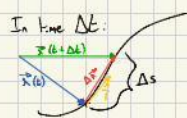
Unit tangent vector: $\hat{T} = \frac{\vec{v}}{|\vec{v}|}$

$$\vec{v} = \frac{d\vec{r}}{dt} = \left(\frac{d\vec{r}}{ds} \right) \left(\frac{ds}{dt} \right) = \hat{T} \frac{ds}{dt}$$

Velocity has: $\begin{cases} \text{direction: tangent to trajectory } \hat{T} \\ \text{length: speed } \frac{ds}{dt} \end{cases}$

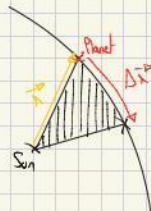


$$\Delta \vec{r} = \hat{T} \Delta s$$



$$\text{Limit as } \Delta t \rightarrow 0 \text{ gives } \frac{d\vec{r}}{dt} = \hat{T} \frac{ds}{dt}$$

$$\frac{\Delta s}{\Delta t} \approx \text{speed} \\ \frac{\Delta \vec{r}}{\Delta t} \approx \hat{T} \frac{\Delta s}{\Delta t}$$



Example Kepler's second law (1609)

Motion of planets is in a plane, and the area is swept out by the line from sun to planet at a constant rate.

Newton later explained this using formula for gravitational attraction. Kepler's law in terms of vectors?

$$\Delta \text{Area} \approx \frac{1}{2} |\vec{r} \times \Delta \vec{r}| \approx \frac{1}{2} |\vec{r} \times \vec{v}| \Delta t \quad \text{The law says } |\vec{r} \times \vec{v}| = \text{constant}$$

2) Plane of motion contains $\vec{r}$ and $\vec{v}$ Direction of $\vec{r} \times \vec{v}$ = normal to the plane of motion

$$\text{Kepler's 2nd Law} \Leftrightarrow \vec{r} \times \vec{v} = \text{constant vector} \Leftrightarrow \frac{d}{dt} (\vec{r} \times \vec{v}) = 0 \Leftrightarrow \underbrace{\frac{d\vec{r}}{dt} \times \vec{v}}_{\vec{v} \times \vec{v} = 0} + \underbrace{\vec{r} \times \frac{d\vec{v}}{dt}}_{\vec{r} \times \vec{a}} = 0 \Leftrightarrow \vec{r} \times \vec{a} = 0 \Leftrightarrow \vec{a} \parallel \vec{r}$$

Kepler's 2nd Law $\Leftrightarrow \vec{a} \parallel \vec{r} \Leftrightarrow \text{gravitational force} \parallel \vec{r}$